

Efficient EGG CANDLER

T.K. HAREENDRAN

Candling eggs while they are incubating is the proven method to test them for fertility. Often, egg candling is done us-

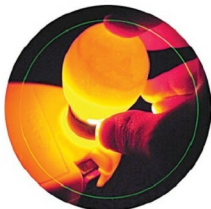


Fig. 1: Typical way of observing the egg through a light source (Courtesy: www.incubatorwarehouse.com)



Fig. 2: Author's prototype with battery

PARTS LIST

Semiconductors:

- D1-D5 - 1N4007 rectifier diode
- ZD1 - 5V, 1W Zener diode
- LED1 - 5mm LED
- LED2 - 1-watt star white LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon, unless stated otherwise):

- R1 - 4.7-ohm, 2W
- R2 - 1-kilo-ohm
- R3 - 390-kilo-ohm
- R4 - 2.2-ohm
- R5 - 22-ohm, 1W

Capacitors:

- C1 - 1 μ F, 400V polyester
- C2 - 100 μ F, 63V electrolytic

Miscellaneous:

- S1 - On/Off switch
- CON1 - 2-pin connector
- BATT.1 - 4V, 500mAh lead-acid battery
- 2-pin terminal connector

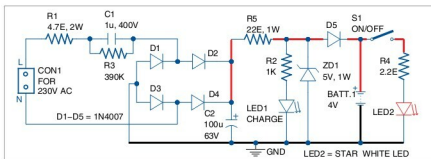


Fig. 3: Circuit diagram of the egg candler

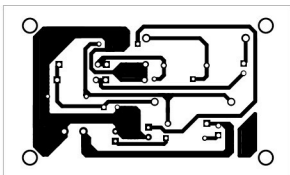


Fig. 4: Actual-size PCB layout of the egg candler

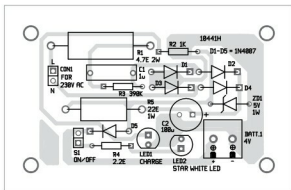


Fig. 5: Components layout for the PCB

ing a light source to channel enough light into the egg in order to determine whether the egg is developing into a chick (see Fig. 1). The light source is usually an incandescent lamp or an LED. Many users prefer an LED as it does not create enough heat to harm the egg.

Here we describe a battery-powered LED egg candler to detect embryonic development or separate eggs with cracked shells. Being battery-powered, the egg candler can be taken anywhere. Also, it is more convenient to use: Just put the egg on the egg base and observe for embryonic growth.

The author's prototype is shown in Fig. 2.

Circuit and working

The circuit (shown in Fig. 3) has a 4V lead-acid SMF battery (BATT.1) and a 1-watt star white LED (LED2) on the right side.

On the left side is a transformerless power supply to charge the

battery. That is, the battery charger is powered by 230V AC mains supply through a transformerless circuit. A 5V Zener diode (ZD1) is used as a pre-regulator to protect low-voltage circuit components from damage by the high-voltage input.

LED1 is the 'charger on' indica-